COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. 8.1 **Install, Configure and Manage Enterprise Availability Monitoring via Ansible**

1. PROCEDURE OUTPUT

1.Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

file structure

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA8$ tree -a
.
├── ansible.cfg
├── .gitignore
├── installing_nagios.yml
├── inventory.yaml
└── roles
    └── nagios
        ├── defaults
        │   └── main.yml
        ├── handlers
        │   └── main.yml
        ├── tasks
        │   ├── build.yml
        │   ├── deps.yml
        │   ├── main.yml
        │   ├── redhat.yml
        │   └── web.yml
        └── vars
            ├── Debian.yml
            └── RedHat.yml
```

- This screenshot shows the tree -a command inside the HOA8 folder. It lists all the files of my project including the hidden .gitignore. The playbook, ansible.cfg and inventory are on top and the nagios role is inside the roles folder with its defaults, handlers, tasks and vars folders.

inventory.yaml

```
 GNU nano 7.2                                          inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas

  children:
    Ubuntu_servers:
      hosts:
        server1:
          ansible_host: 192.168.56.108
        server2:
          ansible_host: 192.168.56.109

    CentOS_servers:
      hosts:
        centOS:
          ansible_host: 192.168.56.104

    linux_servers:
      children:
        Ubuntu_servers:
        CentOS_servers:
```

- This file lists the servers Ansible will manage. It has two groups, Ubuntu servers (server1 and server2) and CentOS servers (centOS) and each with its IP address. The linux servers group combines both groups so one playbook can run on all three machines.

Code

ansible.cfg

```
 GNU nano 7.2                                          ansible.cfg
[defaults]
inventory = inventory.yaml
roles_path = roles
host_key_checking = False
interpreter_python = auto_silent
retry_files_enabled = False

[privilege_escalation]
become = True
become_method = sudo
```

- This is the settings file for Ansible. It tells Ansible to use inventory.yaml and to look for roles in the roles folder. It also turns off SSH host key checking so it does not ask about fingerprints and it uses sudo to run tasks as admin.

installing_nagios.yml (playbook)

```
 GNU nano 7.2                                        installing_nagios.yml
---
- name: Install and Configure Nagios Core across Ubuntu and CentOS
  hosts: linux_servers
  become: true
  roles:
    - nagios
```

- This is the main playbook I run. It targets linux_servers and uses become: true to get admin rights. It only calls the nagios role so all the real work is inside the role.

.gitignore

```
 GNU nano 7.2                                              .gitignore
*.retry
.vault_pass
secrets.yml
*.log
```

- This file tells Git which files not to upload to GitHub. I listed retry files, the vault password file, secrets.yml and log files. This keeps passwords and junk files out of my repository.

roles/nagios/defaults/main.yml

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/defaults$ cat main.yml
---
nagios_version: "4.4.14"
nagios_plugins_version: "2.4.6"

nagios_core_url: "https://assets.nagios.com/downloads/nagioscore/releases/nagios-{{ nagios_version }}.tar.gz"
nagios_plugins_url: "https://github.com/nagios-plugins/nagios-plugins/releases/download/release-{{ nagios_plugins_versi
n }}/nagios-plugins-{{ nagios_plugins_version }}.tar.gz"

nagios_install_dir: /usr/local/nagios
nagios_build_dir: /tmp

nagios_admin_user: charles-vence-espinas
nagios_admin_password: "Password123!"
```

- This file holds the default values of the role. It has the Nagios version (4.4.14), the plugins version (2.4.6), and the download links. It also has the install folder, the build folder, and the username and password for the Nagios web login. Keeping these here means I can change them in one place without editing the tasks.

roles/nagios/handlers/main.yml

```
 GNU nano 7.2                                                main.yml
---
- name: restart apache
  ansible.builtin.service:
    name: "{{ apache_service }}"
    state: restarted

- name: restart nagios
  ansible.builtin.service:
    name: nagios
    state: restarted
```

- Handlers are tasks that only run when another task tells them to. Mine restart Apache and restart Nagios. They run after an install or change so the new settings load, and they do not run when nothing changed.

roles/nagios/tasks/build.yml

```
 GNU nano 7.2                                                build.yml
---
#Nagios Core
- name: Download Nagios Core tarball
  ansible.builtin.get_url:
    url: "{{ nagios_core_url }}"
    dest: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}.tar.gz"
    mode: "0644"

- name: Extract Nagios Core source
  ansible.builtin.unarchive:
    src: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}.tar.gz"
    dest: "{{ nagios_build_dir }}"
    remote_src: true
    creates: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}/configure"

- name: Compile and install Nagios Core
  ansible.builtin.shell: |
    set -e
    ./configure --with-httpd-conf={{ apache_conf_dir }}
    make all
    make install-groups-users
    make install
    make install-daemoninit
    make install-commandmode
    make install-config
    make install-webconf
  args:
    chdir: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}"
```

```
 GNU nano 7.2                                          build.yml
    creates: "{{ nagios_install_dir }}/bin/nagios"
  register: nagios_core_build
  notify:
    - restart apache
    - restart nagios

- name: Reload systemd so the nagios unit is picked up
  ansible.builtin.systemd:
    daemon_reload: true
  when: nagios_core_build is changed

#Nagios Plugins
- name: Download Nagios Plugins tarball
  ansible.builtin.get_url:
    url: "{{ nagios_plugins_url }}"
    dest: "{{ nagios_build_dir }}/nagios-plugins-{{ nagios_plugins_version }}.tar.gz"
    mode: "0644"

- name: Extract Nagios Plugins source
  ansible.builtin.unarchive:
    src: "{{ nagios_build_dir }}/nagios-plugins-{{ nagios_plugins_version }}.tar.gz"
    dest: "{{ nagios_build_dir }}"
    remote_src: true
    creates: "{{ nagios_build_dir }}/nagios-plugins-{{ nagios_plugins_version }}/configure"
```

```
- name: Compile and install Nagios Plugins
  ansible.builtin.shell: |
    set -e
    ./configure
    make
    make install
  args:
    chdir: "{{ nagios_build_dir }}/nagios-plugins-{{ nagios_plugins_version }}"
    creates: "{{ nagios_install_dir }}/libexec/check_ping"
  notify: restart nagios
```

- This file installs Nagios Core and the Nagios Plugins from source. It downloads each tarball, extracts it and then runs configure, make and make install. Each compile step has a creates line so it is skipped if Nagios is already installed. It also reloads systemd so the Nagios service is recognized.

roles/nagios/tasks/deps.yml

```
 GNU nano 7.2                                  deps.yml
---
- name: Update apt cache (Debian family)
  ansible.builtin.apt:
    update_cache: true
    cache_valid_time: 3600
  when: ansible_os_family == "Debian"

- name: Install dnf-plugins-core (RedHat family)
  ansible.builtin.dnf:
    name: dnf-plugins-core
    state: present
  when: ansible_os_family == "RedHat"

- name: Check whether the CRB repository is defined (RedHat family)
  ansible.builtin.shell: dnf repolist --all | grep -q '^crb '
  register: crb_check
  changed_when: false
  failed_when: false
  when: ansible_os_family == "RedHat"

- name: Define the CRB repository if the VM does not have it (RedHat family)
  ansible.builtin.yum_repository:
    name: crb
    description: CentOS Stream $releasever - CRB
    metalink: "https://mirrors.centos.org/metalink?repo=centos-crb-$stream&arch=$basearch&protocol=https,http"
    gpgcheck: true
    gpgkey: file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
    enabled: true
```

```
  when:
    - ansible_os_family == "RedHat"
    - crb_check.rc != 0

- name: Enable the CRB repository if it already exists (RedHat family)
  ansible.builtin.command: dnf config-manager --set-enabled crb
  changed_when: false
  when:
    - ansible_os_family == "RedHat"
    - crb_check.rc == 0

- name: Install build dependencies
  ansible.builtin.package:
    name: "{{ nagios_packages }}"
    state: present
```

- This file installs the packages Nagios needs before it can be built. On Ubuntu it updates the apt cache first. On CentOS it checks whether the CRB repository exists, adds it if it is missing, or enables it if it is there because gd-devel comes from CRB. The last task installs the package list for each OS.

roles/nagios/tasks/main.yml

```
 GNU nano 7.2                                                main.yml
---
- name: Load OS-specific variables
  ansible.builtin.include_vars: "{{ ansible_os_family }}.yml"

- name: Install build dependencies
  ansible.builtin.include_tasks: deps.yml

- name: Open firewall and relax SELinux (RedHat family only)
  ansible.builtin.include_tasks: redhat.yml
  when: ansible_os_family == "RedHat"

- name: Build and install Nagios Core and Plugins
  ansible.builtin.include_tasks: build.yml

- name: Configure Apache, web login and services
  ansible.builtin.include_tasks: web.yml
```

- This is the main task list of the role, and it runs the other files in order. It loads the OS variables, installs the dependencies, runs the CentOS-only tasks, builds Nagios and sets up the web part. The when line makes redhat.yml run only on CentOS.

roles/nagios/tasks/redhat.yml

```
 GNU nano 7.2                                              redhat.yml
---
- name: Gather service facts
  ansible.builtin.service_facts:

- name: Allow HTTP through firewalld
  ansible.posix.firewalld:
    service: http
    permanent: true
    immediate: true
    state: enabled
  when:
    - ansible_facts.services['firewalld.service'] is defined
    - ansible_facts.services['firewalld.service'].state == 'running'

- name: Set SELinux to permissive (lab setting so Nagios CGI scripts work)
  ansible.posix.selinux:
    policy: targeted
    state: permissive
  when: ansible_selinux.status | default('disabled') == 'enabled'
```

- This file has the tasks for CentOS only. It opens HTTP in the firewall so the Nagios page can be reached from a browser. It also sets SELinux to permissive so the Nagios web scripts are not blocked.

roles/nagios/tasks/web.yml

```
 GNU nano 7.2                                                web.yml
---
- name: Enable Apache modules rewrite and cgi (Debian family)
  ansible.builtin.command: "a2enmod {{ item }}"
  loop:
    - rewrite
    - cgi
  register: a2enmod_result
  changed_when: "'already enabled' not in a2enmod_result.stdout"
  notify: restart apache
  when: ansible_os_family == "Debian"

- name: Add the Apache user to the nagios group
  ansible.builtin.user:
    name: "{{ apache_user }}"
    groups: nagios
    append: true
  notify: restart apache

- name: Create the Nagios web admin login
  ansible.builtin.command: >
    htpasswd -bc {{ nagios_install_dir }}/etc/htpasswd.users
    {{ nagios_admin_user }} {{ nagios_admin_password | quote }}
  args:
    creates: "{{ nagios_install_dir }}/etc/htpasswd.users"
  no_log: true
```

```
- name: Enable and start Apache and Nagios
  ansible.builtin.service:
    name: "{{ item }}"
    state: started
    enabled: true
  loop:
    - "{{ apache_service }}"
    - nagios
```

- This file sets up the web side of Nagios. On Ubuntu it enables the Apache rewrite and cgi modules. On both systems it adds the Apache user to the nagios group, creates the web login with htpasswd and then enables and starts Apache and Nagios.

roles/nagios/vars/Debian.yml

```
  GNU nano 7.2                                          Debian.yml
---
apache_service: apache2
apache_user: www-data
apache_conf_dir: /etc/apache2/sites-enabled

nagios_packages:
  - autoconf
  - build-essential
  - gcc
  - make
  - libc6
  - wget
  - tar
  - unzip
  - bc
  - gawk
  - dc
  - gettext
  - apache2
  - apache2-utils
  - php
  - libapache2-mod-php
  - libgd-dev
  - openssl
  - libssl-dev
  - snmp
  - libnet-snmp-perl
```

- This file holds the values for Ubuntu. It sets the Apache service to apache2, the Apache user to www-data, and the config folder. It also has the list of packages to install on Ubuntu.

roles/nagios/vars/RedHat.yml

```
  GNU nano 7.2                                          RedHat.yml
---
apache_service: httpd
apache_user: apache
apache_conf_dir: /etc/httpd/conf.d

nagios_packages:
  - gcc
  - glibc
  - glibc-common
  - make
  - gettext
  - automake
  - autoconf
  - wget
  - tar
  - unzip
  - bc
  - gawk
  - openssl-devel
  - net-snmp
  - net-snmp-utils
  - httpd
  - httpd-tools
  - php
  - gd
  - gd-devel
  - perl
  - python3-libselinux
```

- This is the same idea for CentOS but with different names. The service is httpd, the user is apache, and the config folder is /etc/httpd/conf.d. The package list has the CentOS names, such as gd-devel and httpd-tools.

Output

Workstation

1st attempt: centos failed

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA8$ ansible-galaxy collection install ansi
le.posix
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to reinstall them, consider using `--force`
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA8$ ansible-playbook installing_nagios.yml
--ask-become-pass
BECOME password:

PLAY [Install and Configure Nagios Core across Ubuntu and CentOS] **************

TASK [Gathering Facts] *********************************************************
ok: [server2]
ok: [server1]
ok: [centOS]

TASK [nagios : Load OS-specific variables] *************************************
ok: [server1]
ok: [server2]
ok: [centOS]

TASK [nagios : Install build dependencies] *************************************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/deps.yml for server
1, server2, centOS

TASK [nagios : Update apt cache (Debian family)] *******************************
skipping: [centOS]
changed: [server1]
changed: [server2]
```

```
TASK [nagios : Install dnf-plugins-core (RedHat family)] ***********************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [nagios : Enable the CRB repository which is  needed for gd-devel (RedHat family)] ***
skipping: [server1]
skipping: [server2]
fatal: [centOS]: FAILED! => {"changed": false, "cmd": ["dnf", "config-manager", "--set-enabled", "crb"], "delta": "0:00
00.258045", "end": "2026-09-19 08:55:43.325328", "msg": "non-zero return code", "rc": 1, "start": "2026-09-19 08:55:43.
67283", "stderr": "Error: No matching repo to modify: crb.", "stderr_lines": ["Error: No matching repo to modify: crb."
, "stdout": "", "stdout_lines": []}

TASK [nagios : Install build dependencies] *************************************
changed: [server1]
changed: [server2]

TASK [nagios : Open firewall and relax SELinux (RedHat family only)] ***********
skipping: [server1]
skipping: [server2]

TASK [nagios : Build and install Nagios Core and Plugins] **********************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/build.yml for serv
r1, server2

TASK [nagios : Download Nagios Core tarball] ***********************************
changed: [server2]
changed: [server1]

TASK [nagios : Extract Nagios Core source] *************************************
changed: [server2]
changed: [server1]
```

```
TASK [nagios : Compile and install Nagios Core] ********************************
changed: [server2]
changed: [server1]

TASK [nagios : Reload systemd so the nagios unit is picked up] *****************
ok: [server1]
ok: [server2]

TASK [nagios : Download Nagios Plugins tarball] ********************************
changed: [server2]
changed: [server1]

TASK [nagios : Extract Nagios Plugins source] **********************************
changed: [server2]
changed: [server1]

TASK [nagios : Compile and install Nagios Plugins] *****************************
changed: [server2]
changed: [server1]

TASK [nagios : Configure Apache, web login and services] ***********************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/web.yml for server1
, server2

TASK [nagios : Enable Apache modules rewrite and cgi (Debian family)] **********
changed: [server2] => (item=rewrite)
changed: [server1] => (item=rewrite)
changed: [server2] => (item=cgi)
changed: [server1] => (item=cgi)
```

```
TASK [nagios : Add the Apache user to the nagios group] ************************
changed: [server2]
changed: [server1]

TASK [nagios : Create the Nagios web admin login] ******************************
changed: [server2]
changed: [server1]

TASK [nagios : Enable and start Apache and Nagios] *****************************
ok: [server1] => (item=apache2)
ok: [server2] => (item=apache2)
changed: [server1] => (item=nagios)
changed: [server2] => (item=nagios)

RUNNING HANDLER [nagios : restart apache] **************************************
changed: [server1]
changed: [server2]

RUNNING HANDLER [nagios : restart nagios] **************************************
changed: [server1]
changed: [server2]

PLAY RECAP *********************************************************************
centOS                     : ok=4    changed=0    unreachable=0    failed=1    skipped=1    rescued=0    ignored=0
server1                    : ok=20   changed=14   unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server2                    : ok=20   changed=14   unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

- The playbook failed on CentOS because the CRB repository was missing and it is needed to install gd-devel. To fix this, I edited deps.yml so the playbook first checks if the CRB repo exists. If it is missing, the playbook adds it, and if it exists, the playbook enables it. After that, I ran the playbook again and the installation continued without errors.

2nd attempt:

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/HOA8$ ansible-playbook installing_nagios.yml
--ask-become-pass
BECOME password:

PLAY [Install and Configure Nagios Core across Ubuntu and CentOS] ***************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server2]
ok: [centOS]
ok: [server1]

TASK [nagios : Load OS-specific variables] *************************************************************************
ok: [server1]
ok: [server2]
ok: [centOS]

TASK [nagios : Install build dependencies] *************************************************************************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/deps.yml for server
1, server2, centOS

TASK [nagios : Update apt cache (Debian family)] *******************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

TASK [nagios : Install dnf-plugins-core (RedHat family)] ***********************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]
```

```
TASK [nagios : Check whether the CRB repository is defined (RedHat family)] ****************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [nagios : Define the CRB repository if the VM does not have it (RedHat family)] ******************************
skipping: [server1]
skipping: [server2]
changed: [centOS]

TASK [nagios : Enable the CRB repository if it already exists (RedHat family)] *************************************
skipping: [server1]
skipping: [server2]
skipping: [centOS]

TASK [nagios : Install build dependencies] *************************************************************************
ok: [server2]
ok: [server1]
changed: [centOS]

TASK [nagios : Open firewall and relax SELinux (RedHat family only)] ***********************************************
skipping: [server1]
skipping: [server2]
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/redhat.yml for cent
OS

TASK [nagios : Gather service facts] *******************************************************************************
ok: [centOS]
```

```
TASK [nagios : Allow HTTP through firewalld] ***********************************************************
changed: [centOS]

TASK [nagios : Set SELinux to permissive (lab setting so Nagios CGI scripts work)] *********************
changed: [centOS]

TASK [nagios : Build and install Nagios Core and Plugins] **********************************************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/build.yml for serve
r1, server2, centOS

TASK [nagios : Download Nagios Core tarball] ***********************************************************
ok: [server1]
changed: [centOS]
ok: [server2]

TASK [nagios : Extract Nagios Core source] *************************************************************
skipping: [server2]
changed: [centOS]
skipping: [server1]

TASK [nagios : Compile and install Nagios Core] ********************************************************
ok: [server1]
ok: [server2]
changed: [centOS]

TASK [nagios : Reload systemd so the nagios unit is picked up] *****************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]
```

```
TASK [nagios : Download Nagios Plugins tarball] ********************************************************
ok: [server2]
changed: [centOS]
ok: [server1]

TASK [nagios : Extract Nagios Plugins source] **********************************************************
skipping: [server1]
skipping: [server2]
changed: [centOS]

TASK [nagios : Compile and install Nagios Plugins] *****************************************************
ok: [server2]
ok: [server1]
changed: [centOS]

TASK [nagios : Configure Apache, web login and services] ***********************************************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/HOA8/roles/nagios/tasks/web.yml for server1
, server2, centOS

TASK [nagios : Enable Apache modules rewrite and cgi (Debian family)] **********************************
skipping: [centOS] => (item=rewrite)
skipping: [centOS] => (item=cgi)
skipping: [centOS]
ok: [server2] => (item=rewrite)
ok: [server2] => (item=cgi)
ok: [server1] => (item=rewrite)
ok: [server1] => (item=cgi)
```

```
TASK [nagios : Add the Apache user to the nagios group] ************************************************************
ok: [server1]
ok: [server2]
changed: [centOS]

TASK [nagios : Create the Nagios web admin login] ******************************************************************
ok: [server2]
changed: [centOS]
ok: [server1]

TASK [nagios : Enable and start Apache and Nagios] *****************************************************************
ok: [server1] => (item=apache2)
ok: [server2] => (item=apache2)
ok: [centOS] => (item=httpd)
ok: [server1] => (item=nagios)
ok: [server2] => (item=nagios)
changed: [centOS] => (item=nagios)

RUNNING HANDLER [nagios : restart apache] **************************************************************************
changed: [centOS]

RUNNING HANDLER [nagios : restart nagios] **************************************************************************
changed: [centOS]

PLAY RECAP *********************************************************************************************************
centOS                     : ok=25   changed=15   unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server1                    : ok=15   changed=0    unreachable=0    failed=0    skipped=8    rescued=0    ignored=0
server2                    : ok=15   changed=0    unreachable=0    failed=0    skipped=8    rescued=0    ignored=0
```

- After I fixed deps.yml, I ran the playbook again. The new tasks checked for the CRB repo and added it on CentOS, then the CentOS install went through until the end. The Ubuntu servers only showed ok because Nagios was already installed. The recap shows failed=0 for all three servers.

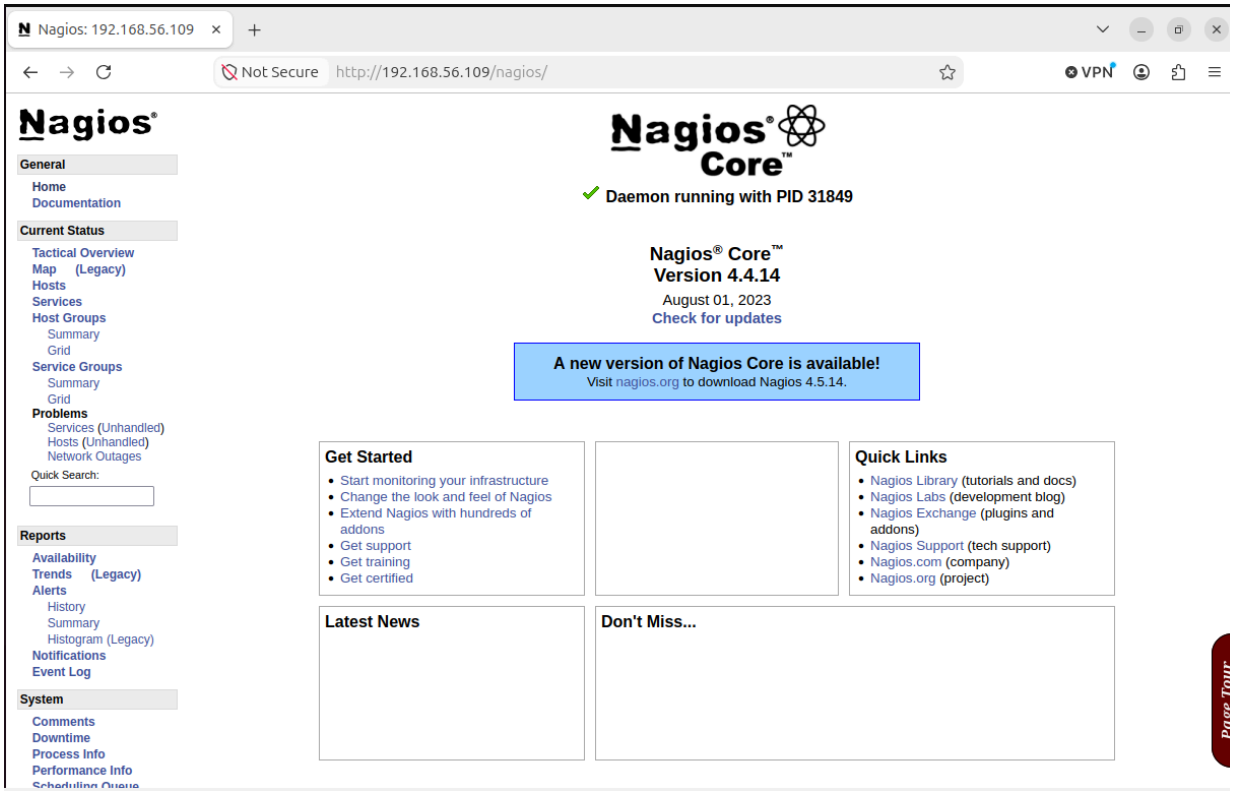
Server 1

```
charles-vence-espinas@server1:~$ systemctl status nagios
● nagios.service - Nagios Core 4.4.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: e>
     Active: active (running) since Sat 2026-09-19 12:58:55 UTC; 15min ago
       Docs: https://www.nagios.org/documentation
    Process: 32372 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagi>
    Process: 32375 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/>
   Main PID: 32376 (nagios)
      Tasks: 6 (limit: 1374)
     Memory: 5.6M (peak: 8.6M)
        CPU: 246ms
     CGroup: /system.slice/nagios.service
             ├─32376 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
             ├─32378 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─32379 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─32380 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─32381 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             └─32382 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
```



Server 2

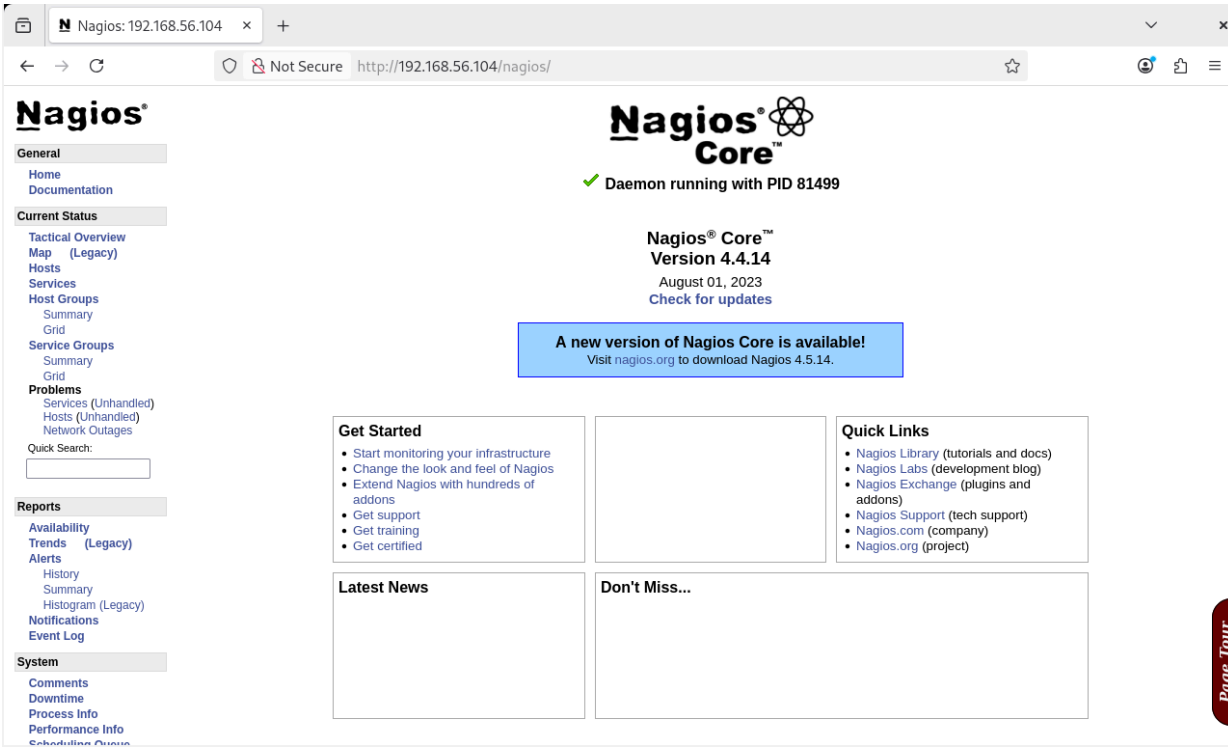
```
charles-vence-espinas@server2:~$ systemctl status nagios
● nagios.service - Nagios Core 4.4.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: e>
     Active: active (running) since Sat 2026-09-19 12:58:55 UTC; 50min ago
       Docs: https://www.nagios.org/documentation
   Main PID: 31849 (nagios)
      Tasks: 6 (limit: 1374)
     Memory: 9.1M (peak: 11.2M)
        CPU: 929ms
     CGroup: /system.slice/nagios.service
             ├─31849 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
             ├─31850 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─31851 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─31852 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─31853 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             └─31854 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
```


Nagios: 192.168.56.109
Not Secure
http://192.168.56.109/nagios/
VPN
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map (Legacy)
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends (Legacy)
Alerts
History
Summary
Histogram (Legacy)
Notifications
Event Log
System
Comments
Downtime
Process Info
Performance Info
Scheduling Queue
Nagios® Core™
Daemon running with PID 31849
Nagios® Core™
Version 4.4.14
August 01, 2023
Check for updates
A new version of Nagios Core is available!
Visit nagios.org to download Nagios 4.5.14.
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CentOS

```
[charles-vence-espinas@centOS ~]$ systemctl status nagios
● nagios.service - Nagios Core 4.4.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: d>
     Active: active (running) since Sat 2026-09-19 09:11:22 EDT; 44min ago
       Docs: https://www.nagios.org/documentation
    Process: 81497 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagi>
    Process: 81498 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/>
   Main PID: 81499 (nagios)
      Tasks: 8 (limit: 10189)
     Memory: 36.5M (peak: 40.8M)
        CPU: 801ms
     CGroup: /system.slice/nagios.service
             ├─81499 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
             ├─81500 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─81501 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─81502 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─81503 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─81504 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             ├─81505 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
             └─81506 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>

Sep 19 09:11:22 centOS nagios[81499]: wproc: Registry request: name=Core Worker>
lines 1-21...skipping...
● nagios.service - Nagios Core 4.4.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-09-19 09:11:22 EDT; 44min ago
       Docs: https://www.nagios.org/documentation
    Process: 81497 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg (code=exited, status=0/SUCCESS)
    Process: 81498 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg (code=exited, status=0/SUCCESS)
   Main PID: 81499 (nagios)
      Tasks: 8 (limit: 10189)
     Memory: 36.5M (peak: 40.8M)
        CPU: 801ms
     CGroup: /system.slice/nagios.service
             ├─81499 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             ├─81500 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─81501 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─81502 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─81503 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─81504 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─81505 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             └─81506 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
```



- For each server, I show the systemctl status nagios command which reports that the service is active

(running). Next to it is the Nagios web page (server1 at 192.168.56.108, server2 at 192.168.56.109 and CentOS at 192.168.56.104). Each page says Nagios Core 4.4.14 with "Daemon running" so the install worked on Ubuntu and CentOS.

Github repository link:
https://github.com/charlesespinas/CPE212_CharlesVenceEspinas

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What are the benefits of having an availability monitoring tool?
   - The key benefits of having an availability monitoring tool are that it checks all the time whether the servers and apps are up and reachable. It alerts the admin right away when something goes down so the problem can be fixed before the users notice it. This lowers the downtime which protects the company's income and reputation. It also keeps the records of the past outages which helps the team find problems and plan for the future. Overall, the admin does not need to check every server by hand so it saves time and efforts.

Conclusion

In conclusion, I learned how to install Nagios on both ubuntu and centos using one ansible playbook and role. Using the variable files for each Os made the same tasks work on two different systems. I also learned that errors can happen likein my case which is missing CRB repository on the centos and that reading the error messages helps me fix them. Handlers and the creates option made the playbook cleaner because it does not repeat work tha is already done. In the end, nagios was running on all three servers and I learned how ansible can make the server setup faster and easier.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concept**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use this tool as an assistant to help me when there's part of the code that I have not familiar yet and I will make it explain in detailed and then I will write my own code and I will let the tool evaluate if it is correct**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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